

26-2 EtherChannel Troubleshooting - Answer Key

In this lab you will troubleshoot EtherChannel.

EtherChannel Troubleshooting

The junior network administrator was tasked with configuring EtherChannel links over the weekend:

1. LACP between the FastEthernet 0/23 and 0/24 interfaces on Acc3 and CD1.
2. LACP between the FastEthernet 0/21 and 0/22 interfaces on Acc3 and CD2.
3. PAgP between the FastEthernet 0/23 and 0/24 interfaces on Acc4 and CD2.
4. PAgP between the FastEthernet 0/21 and 0/22 interfaces on Acc4 and CD1.
5. Static EtherChannel between the GigabitEthernet 0/1 and 0/2 interfaces on CD1 and CD2.

You have discovered there are issues with the links on Monday morning and there is no connectivity between PCs on different access layer switches.

Note that Packet Tracer has a bug which may cause 'show etherchannel' commands to report the Port Channels as up even though they are misconfigured.

Your task is to troubleshoot and correct the configuration in order to meet the requirements above and restore connectivity.

You may need to flap the interfaces with a 'shutdown/no shutdown' to get the EtherChannels to come up after making changes.

There is not a set order of actions to troubleshoot this scenario. Troubleshooting in a logical fashion will however make it easier and quicker. This is how I would do it.

Then check the other side on CD1.

```
CD1#sh run
!  
interface Port-channel 1  
description Link to Acc3  
switchport trunk native vlan 199  
switchport mode trunk  
!  
interface FastEthernet0/23  
switchport trunk native vlan 199  
switchport mode trunk  
channel-group 1 mode passive  
!  
interface FastEthernet0/24  
switchport trunk native vlan 199  
switchport mode trunk  
channel-group 1 mode passive  
!  
Truncated
```

We can see that the member ports on both sides have been set to passive. LACP will form an EtherChannel if both sides are set to active, or if one side is active and the other is passive, but not if both sides are passive. Best practice is to set both sides to active to minimize the chance of a mismatch. Correct the configuration, then flap the interfaces to reset them.

```
Acc3(config)#int range f0/23 - 24  
Acc3(config-if-range)#channel-group 1 mode active  
Acc3(config-if-range)#shutdown  
Acc3(config-if-range)#no shutdown  
!  
Acc3(config)#int po1  
Acc3(config-if)#shutdown  
Acc3(config-if)#no shutdown  
  
CD1(config)#int range f0/23 - 24  
CD1(config-if-range)#channel-group 1 mode active  
CD1(config-if-range)#shutdown  
CD1(config-if-range)#no shutdown  
!  
CD1(config)#int po1  
CD1(config-if)#shutdown  
CD1(config-if)#no shutdown
```

Verify the port channel comes up. (You may need to flap the FastEthernet and Port Channel interfaces with a 'shutdown' then 'no shutdown' to get the EtherChannel to come up.)

```
Acc3#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 2
Number of aggregators: 2
```

Group Port-channel Protocol Ports

```
-----+-----+-----+-----
1 Po1(SU) LACP Fa0/23(P) Fa0/24(P)
2 Po2(SD) LACP Fa0/21(I) Fa0/22(I)
```

2. LACP between the FastEthernet 0/21 and 0/22 interfaces on Acc3 and CD2.

First verify the EtherChannel is down.

```
Acc3#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 2
Number of aggregators: 2
```

Group Port-channel Protocol Ports

```
-----+-----+-----+-----
1 Po1(SU) LACP Fa0/23(P) Fa0/24(P)
2 Po2(SD) LACP Fa0/21(I) Fa0/22(I)
```

```
Acc3#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 2
Number of aggregators: 2
```

```

-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 Po1 (SU) LACP Fa0/23 (P) Fa0/24 (P)
2 Po2 (SU) LACP Fa0/21 (P) Fa0/22 (P)

```

```
Acc3#show run
!
interface Port-channel 2
description Link to CD2
switchport trunk native vlan 199
switchport mode trunk
!
interface FastEthernet0/21
switchport trunk native vlan 199
switchport mode trunk
speed 10
channel-group 2 mode active
!
interface FastEthernet0/22
switchport trunk native vlan 199
switchport mode trunk
speed 10
channel-group 2 mode active
!
Truncated
```

Then check the other side on CD2.

```
CD2#show run
!  
interface Port-channel 2  
description Link to Acc3  
switchport trunk native vlan 199  
switchport mode trunk  
!  
interface FastEthernet0/21  
switchport trunk native vlan 199  
switchport mode trunk  
channel-group 2 mode active  
!  
interface FastEthernet0/22  
switchport trunk native vlan 199  
switchport mode trunk  
channel-group 2 mode active  
!  
Truncated
```

We can see that there is a speed mismatch. The ports on Acc3 are set to 10 Mbps whereas the ports on CD2 are running at the default 100 Mbps for FastEthernet interfaces. An EtherChannel will not be formed if the interface settings do not match on both sides. Set the speed on Acc3 back to the default auto setting.

```
Acc3(config)#int range f0/21 - 22  
Acc3(config-if-range)#no speed 10
```

Verify the port channel comes up. (You may need to flap the FastEthernet and Port Channel interfaces with a 'shutdown' then 'no shutdown' to get the EtherChannel to come up.)

```
Acc3(config)#int range f0/21 - 22
Acc3(config-if-range)#shutdown
Acc3(config-if-range)#no shutdown
```

```
Acc3#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
```

```
Number of channel-groups in use: 2
Number of aggregators: 2
```

```
Group Port-channel Protocol Ports
```

```
-----+-----+-----+-----
```

```
1 Po1(SU) LACP Fa0/23(P) Fa0/24(P)
2 Po2(SU) LACP Fa0/21(P) Fa0/22(P)
```

3. PAgP between the FastEthernet 0/23 and 0/24 interfaces on Acc4 and CD2.

First verify the EtherChannel is down.

Note that Packet Tracer has a bug which *may* cause the 'show etherchannel summary' command to report the Port Channel as up even though it is misconfigured.

```
Acc4#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 2
Number of aggregators: 2
```

```
Group Port-channel Protocol Ports
```

```
-----+-----+-----+-----
```

```
1 Po1(SD) PAgP Fa0/23(I) Fa0/24(I)
2 Po2(SD) PAgP Fa0/21(I) Fa0/22(I)
```

Check the configuration on the port channel and member interfaces. We'll look at Acc4 first.

```
Acc4#show run
!
interface Port-channel 1
switchport trunk native vlan 199
switchport mode trunk
!
interface FastEthernet0/23
switchport trunk native vlan 199
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/24
switchport trunk native vlan 199
switchport mode trunk
channel-group 1 mode desirable
!
Truncated
```


Then check the other side on CD2.

```
CD2#show run
!
interface Port-channel 1
switchport trunk native vlan 199
switchport mode trunk
!
interface FastEthernet0/23
switchport trunk native vlan 199
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/24
switchport trunk native vlan 199
switchport mode trunk
channel-group 1 mode active
!
Truncated
```

We can see that there is an EtherChannel type mismatch. The ports on Acc4 are set to PAgP desirable whereas the ports on CD2 are set to LACP active. The requirement was for PAgP to be used between Acc4 and CD2 so set the ports on CD2 to PAgP desirable.

```
CD2(config)#int range f0/23 - 24
CD2(config-if-range)#no channel-group 1 mode active
CD2(config-if-range)#channel-group 1 mode desirable
```

Verify the port channel comes up.

```
Acc4#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 2
Number of aggregators: 2
```

```
Group Port-channel Protocol Ports
```

```
-----+-----+-----+-----
1 Po1(SU) PAgP Fa0/23(P) Fa0/24(P)
2 Po2(SD) PAgP Fa0/21(I) Fa0/22(I)
```

4. PAgP between the FastEthernet 0/21 and 0/22 interfaces on Acc4 and CD1.

First verify the EtherChannel is down.

```
Acc4#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 2
Number of aggregators: 2
```

```
Group Port-channel Protocol Ports
-----+-----+-----+-----
```

```
1 Po1(SU) PAgP Fa0/23(P) Fa0/24(P)
2 Po2(SD) PAgP Fa0/21(I) Fa0/22(I)
```

Note that Packet Tracer has a bug which *may* cause the 'show etherchannel summary' command to report the Port Channel as up as shown below even though it is misconfigured.

```
Acc4#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 2
Number of aggregators: 2
```

```
Group Port-channel Protocol Ports
-----+-----+-----+-----
```

```
1 Po1(SU) PAgP Fa0/23(P) Fa0/24(P)
2 Po2(SU) PAgP Fa0/21(P) Fa0/22(P)
```

In that case 'show spanning-tree' should still show the Port Channel interface is *TYPE_Inc (inconsistent) for VLAN 1 and not available for the PC VLAN 10.

```
Acc4#show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
Root ID    Priority    24577
           Address    0090.0CA0.3902
           Cost        9
           Port        28 (Port-channel2)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0060.708A.D564
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/24	Desg	FWD	19	128.24	P2p
Fa0/23	Desg	FWD	19	128.23	P2p
Po1	Desg	FWD	9	128.27	Shr
Po2	Root	BKN*9		128.28	Shr *TYPE_Inc

```
VLAN0010
```

```
Spanning tree enabled protocol ieee
Root ID    Priority    24586
           Address    0090.0CA0.3902
           Cost        28
           Port        27 (Port-channel1)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    32778 (priority 32768 sys-id-ext 10)
           Address    0060.708A.D564
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/24	Desg	FWD	19	128.24	P2p
Fa0/23	Desg	FWD	19	128.23	P2p
Po1	Root	FWD	9	128.27	Shr

Check the configuration on the port channel and member interfaces. We'll look at Acc4 first.

```
Acc4#sh run
!  
interface Port-channel 2  
switchport trunk native vlan 199  
switchport mode access  
!  
interface FastEthernet0/21  
switchport trunk native vlan 199  
switchport mode access  
channel-group 2 mode desirable  
!  
interface FastEthernet0/22  
switchport trunk native vlan 199  
switchport mode access  
channel-group 2 mode desirable  
!  
Truncated
```

Then check the other side on CD1.

```
CD1#sh run
!  
interface Port-channel 2  
switchport trunk native vlan 199  
switchport mode trunk  
!  
interface FastEthernet0/21  
switchport trunk native vlan 199  
switchport mode trunk  
channel-group 2 mode desirable  
!  
interface FastEthernet0/22  
switchport trunk native vlan 199  
switchport mode trunk  
channel-group 2 mode desirable  
Truncated
```

We can see that there is a switchport type mismatch. The ports on Acc4 are set as access ports whereas the ports on CD1 are set to trunk ports. Set the ports on Acc4 as trunk ports.

```
Acc4(config)#int range f0/21 - 22
Acc4(config-if-range)#switchport mode trunk
Acc4(config-if-range)#exit
Acc4(config)#int port-channel 2
Acc4(config-if)#switchport mode trunk
```

Verify the port channel comes up.

```
Acc4#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 2
Number of aggregators: 2
```

```
Group Port-channel Protocol Ports
```

```
-----+-----+-----+-----
1 Po1(SU) PAgP Fa0/23(P) Fa0/24(P)
2 Po2(SU) PAgP Fa0/21(P) Fa0/22(P)
```

5. Static EtherChannel between the GigabitEthernet 0/1 and 0/2 interfaces on CD1 and CD2.

First verify the EtherChannel is down.

Note that Packet Tracer has a bug which *may* cause the 'show etherchannel summary' command to report the Port Channel as up even though it is misconfigured.

```
CD1#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 3
Number of aggregators: 3
```

```
Group Port-channel Protocol Ports
```

```
-----+-----+-----+-----+
1 Po1(SU) LACP Fa0/23(P) Fa0/24(P)
2 Po2(SU) PAgP Fa0/21(P) Fa0/22(P)
3 Po3(SD) - Gig0/1(D) Gig0/2(D)
```

Check the configuration on the port channel and member interfaces. We'll look at CD1 first.

```
CD1#sh run
!
interface Port-channel 3
switchport trunk native vlan 199
switchport mode trunk
!
interface GigabitEthernet0/1
switchport trunk native vlan 199
switchport mode trunk
channel-group 3 mode on
!
interface GigabitEthernet0/2
switchport trunk native vlan 199
switchport mode trunk
channel-group 3 mode on
!
Truncated
```

Then check the other side on CD2.

```
CD2#show run
!  
interface Port-channel 3  
switchport trunk native vlan 199  
switchport mode trunk  
!  
interface GigabitEthernet0/1  
switchport trunk native vlan 199  
switchport mode trunk  
channel-group 3 mode on  
shutdown  
!  
interface GigabitEthernet0/2  
switchport trunk native vlan 199  
switchport mode trunk  
channel-group 3 mode on  
shutdown  
!  
Truncated
```

We can see that the interfaces on CD2 are administratively shutdown so of course the EtherChannel will not form. Bring the interfaces up.

```
CD2(config)#interface range gig0/1 - 2  
CD2(config-if-range)#no shutdown
```

```
CD2#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
```

```

Group Port-channel Protocol Ports
-----+-----+-----+-----
1 Po1(SU) PAgP Fa0/23(P) Fa0/24(P)
2 Po2(SU) LACP Fa0/21(P) Fa0/22(P)
3 Po3(SU) - Gig0/1(P) Gig0/2(P)

```

```
C:\>ping 10.10.10.12

Pinging 10.10.10.12 with 32 bytes of data:

Reply from 10.10.10.12: bytes=32 time=58ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time=16ms TTL=128
Reply from 10.10.10.12: bytes=32 time=1ms TTL=128

Ping statistics for 10.10.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 58ms, Average = 18ms
```